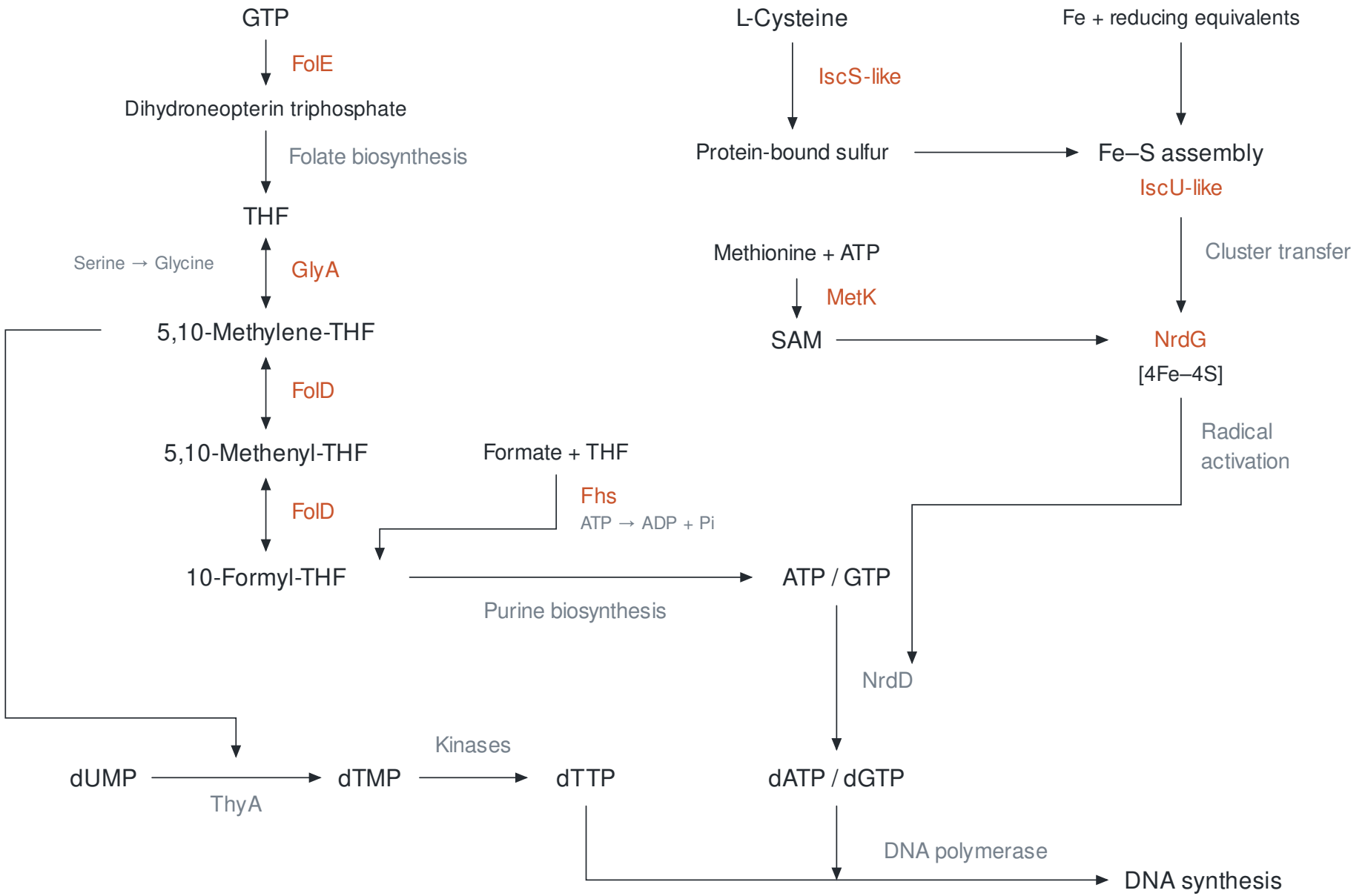
GTP
FolE
Dihydroneopterin triphosphate
Folate biosynthesis
THF
Serine → Glycine
GlyA
5,10-Methylene-THF
FolD
5,10-Methenyl-THF
FolD
10-Formyl-THF
Formate + THF
Fhs
ATP → ADP + Pi
Purine biosynthesis
ATP / GTP
L-Cysteine
IscS-like
Protein-bound sulfur
Fe + reducing equivalents
Fe–S assembly
IscU-like
Cluster transfer
Methionine + ATP
MetK
SAM
NrdG
[4Fe–4S]
Radical activation
NrdD
dUMP
ThyA
dTMP
Kinases
dTTP
dATP / dGTP
DNA polymerase
DNA synthesis